

Profile

Research focuses on embodied intelligence policy learning, vision-conditioned action generation, and shared autonomy. **As a core member of the Peking University Open Robot Lab during his MSc**, he developed a strong foundation in real-robot systems. End-to-end expertise spans large-scale simulation data generation, generative-model training (Transformer/CVAE/Diffusion), ROS 2 deployment on physical robots, and human-subject studies, with a focus on robust sim-to-real transfer. Led projects in vision-guided SE(3) trajectory generation and shared control, as well as volumetric 6-DoF grasp-pose detection. Publications include **IJRR, RA-L, ICRA, CoRL, and RSS. A T-RO paper is under review.**

Education

Cornell University | Visiting Scholar | 2026.09 - Present

GRILL Lab, Supervisor: **Prof. Fang Kuan**

KU Leuven | Ph.D. Student in Robotics | 2023.02 - Present

Robotics Research Group & Processing of Speech and Image, Supervisor: **Prof. Renaud Detry**

Peking University | Applied Computer Science | 2019.09–2022.07

Peking University Open Robot Lab (PKU-HRI), Supervisor: Prof. Hong Liu, GPA : 3.6/4.0

South China University of Technology | Mechanical Engineer (Honors Innovation Program) | 2015.09–2019.07

GPA : 3.74/4.0 (top 5%)

Project

Robot Imitation Learning and Shared Control for Manipulation and Navigation | Belgium | 2023–2026

- Proposed Robot Trajectron (RT), an RNN-CVAE-GMM trajectory predictor that infers user intent from environmental context and robot trajectory history, then guides the user toward a target object for 2D tabletop grasping assistance. **Accepted at ICRA 2024.**
- Extended this work to 2D navigation with RT-V2, using model-based reinforcement learning (MBRL) and model predictive path integral control (MPPI) to improve obstacle avoidance. Formulated assistance as Bayesian posterior inference over an imitation-learning policy for safe, seamless assistance. **Accepted at IJRR.**
- Proposed RT-V3 for SE(3) manipulation. A Transformer fuses a single-view point cloud, candidate SE(3) grasp poses, and robot state history to predict multimodal 6-DoF future action trajectories. It combines the generative policy with real-time user commands through Bayesian posterior inference for closed-loop posterior updates and robotic assistance. **Submitted to T-RO.**

Large-Scale 6-DoF Grasp Learning and Sim-to-Real Deployment | Belgium | 2023–2026

- Developed IGD, a 6-DoF grasping model based on **implicit neural representations** and **diffusion models**. It predicts grasp quality, position, orientation, and gripper width from a single-view depth observation. **Accepted at CoRL 2024.**
- Proposed a **C4-equivariant voxel encoder** in IGD and proposed EquiIGD, which explicitly exploits rotational symmetry to improve data efficiency and robustness to viewpoint variation and depth noise. **Submitted to IJRR.**

Underwater Robotics | Peking University–Peng Cheng Laboratory Collaborative Project | 2019–2022

- Used first-stage prediction errors to supervise the second stage of a two-stage detector, enabling hard-example mining and training a fast, accurate underwater seafood detector. **Accepted by Neurocomputing (340+ citations).**
- Used style transfer for data augmentation across water conditions and contrastive learning to extract water-condition-invariant backbone features, enabling robust cross-domain underwater detection. **Accepted by Neurocomputing (90+ citations) and ICIP 2020 (140+ citations).**

Intern

PlatoOCR | ByteDance, Shanghai | Jul. 2021–Oct. 2021

Developed text detection and recognition under weak supervision. To address the lack of character-level labels in production data, proposed a semi-supervised framework with paragraph-to-character hierarchical consistency constraints. Combined real paragraph-level annotations with synthetic character data and designed high-confidence pseudo-label self-training, enabling end-to-end character detection without fine-grained labels. Addressed the domain gap between synthetic and real data, substantially improving generalization in real-world settings.

Publication

- **Song, P.**, Li, P., Aertbeliën, E. and Detry, R.. Robot trajectron: Trajectory prediction-based shared control for robot manipulation. *ICRA 2024*. (Citation: 30+)
- **Song, P.**, Li, P. and Detry, R.. Implicit grasp diffusion: Bridging the gap between dense prediction and sampling-based grasping. *CORL 2024*.
- **Song, P.**, Du, Y., Saussus, O., De Schrijver, S., Caprara, I., Janssen, P. and Detry, R.. Robot Trajectron V2: A Probabilistic Shared Control Framework for Navigation. *IJRR 2026*.
 - 2025 IROS SASA workshop Best Poster Award 3rd place
- **Song, P.**, Hu, Y., Li, P. and Detry, R.. Equivariant Volumetric Grasping. **Submitted to IJRR**.
- **Song, P.**, Fu, Z., Hu, Y. and Detry, R.. Assistron: Bayesian Shared Autonomy with Off-the-shelf Vision-Language-Action Models. **Submitted to ICRA 2027**.
- **Song, P.**, Fu, Z., Hu, Y. and Detry, R.. Robot Trajectron V3: A Probabilistic Shared Control Framework for SE(3) Manipulation. **Submitted to TRO**.
- Du, Y., **Song, P.**, Hu, Y. and Detry, R.. ELVIS: Ensemble-Calibrated Latent Imagination for Long-Horizon Visual MPC. *RSS 2026*.
- Hu, Y., **Song, P.**, Wen, K. and Detry, R.. Mini Diffuser: Fast Multi-Task Diffusion Policy Training Using Two-Level Mini-Batches. *RAL 2026*.
- **Song, P.**, Li, P., Dai, L., Wang, T. and Chen, Z.. Boosting R-CNN: Reweighting R-CNN samples by RPN’s error for underwater object detection. *Neurocomputing 2023*. (**JCR Q1**, Citation: 330+)
- Chen, Y., **Song, P.**, Liu, H., Dai, L., Zhang, X., Ding, R. and Li, S., 2023. Achieving domain generalization for underwater object detection by domain mixup and contrastive learning. *Neurocomputing 2023*. (**JCR Q1**, Citation: 90+) Full publications can be found in [this link](#).

Award

- 2020 Peking University Research Progress Award
- 2018 Silver Award, SCUT Challenge Cup Student Innovation and Entrepreneurship Competition (Provincial Level, Project Lead)
- 2018 Second-Class University Scholarship, SCUT; 2017 Third-Class University Scholarship, SCUT
- 2016 Koide Scholarship, SCUT (Elite Scholarship, Top 1%)

Skill

Theory	Diffusion model, CVAE, Transformer, IL/RL, Grasping, Probabilistic Model
Programming	python, ROS2, C/C++, MATLAB, pytorch, opencv
Software	Pybullet, Mujoco, IssacSim, Office, PS, CAD
Language	English (IELTS: 7), Japanese (JPLT N1), Dutch (A2)